

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

Phase Maturity Plan

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1 Purpose

This plan extends the design dependency map (CFU-000-PM-DEP-001) across the four phases of execution: FEL 1 (appraise), FEL 2 (select), FEL 3 (define / FEED) and detailed design. Most activities run in every phase. What changes is how mature each one must be at each gate, so the plan is a matrix of target maturity by activity and gate, plus the maturity each link needs.

Use it to:

- see what each gate requires, and what is still short;
- plan each phase as the steps that take activities from one level to the next, in the order the links allow;
- check that the gate targets are consistent (no gate asks for a result whose inputs are not mature enough).

2 Maturity scale

Level	Name	Meaning
0	Not started	—
1	Concept	Sketches, factored or assumed data, option lists
2	Preliminary	Issued for review (rev A); enough to estimate and screen
3	Defined	Issued for design, HAZOP or purchase; the frozen basis for the next phase
4	Final	Issued for construction, approved or certified; one-off studies and decisions when complete

3 Phases and gates

Phase	Gate	Estimate	Purpose	Activities needed
FEL 1 · Appraise	Gate 1 · opportunity confirmed	Class 5	Confirm there is a business case and a technically feasible concept worth developing.	13
FEL 2 · Select	Gate 2 · concept selected	Class 4	Compare the alternatives and select one configuration, technology and site to take forward.	34
FEL 3 · Define (FEED)	Gate 3 · final investment decision	Class 3	Define the selected concept well enough to fund it: HAZOP'd P&IDs, datasheets, long-lead orders.	54
Detailed design	Gate 4 · issued for construction	Class 2	Complete the design with certified vendor data and issue everything needed to build.	61

The plan has 61 activities (45 from the dependency map, 16 that only appear once the phases are modelled), 178 input links, 15 iteration loops and 244 maturity steps.

4 Target matrix

Target level at each gate. "Now" is the level this package has reached.

ID	Activity	Discipline	FEL 1	FEL 2	FEL 3	DD	Now
FEL-01	Business case, capacity and product slate	Project	3	4	4	4	4*

ID	Activity	Discipline	FEL 1	FEL 2	FEL 3	DD	Now
PRJ-01	Design basis (BOD)	Project	2	3	4	4	4
PRJ-03	Document register and portal	Project	1	2	3	4	3
PRJ-05	Execution plan and contracting strategy	Project	1	2	4	4	0
SEL-04	Permitting and environmental basis	Project	.	2	3	4	0
PRJ-04	PE review, seal and IFC issue	Project	.	.	.	4	0
FEL-02	Configuration options and screening	Process	4	4	4	4	4*
PRC-01	Heat and material balance	Process	1	2	3	4	2
PRC-02	Process equipment sizing	Process	1	2	3	4	3
PRC-05	Block flow diagram	Process	1	3	4	4	4
PRC-06	Process flow diagrams	Process	1	2	3	4	3
PRJ-02	Crude assay characterisation	Process	1	2	4	4	2
PRC-00	Rigorous simulation and model calibration	Process	.	1	4	4	0
PRC-03	Relief load and PSV sizing (unit)	Process	.	1	3	4	3
PRC-04	Control loops and SIF definition	Process	.	1	3	4	3
PRC-07	Process design report	Process	.	1	4	4	4
PRC-08	P&IDs, line list, instrument index	Process	.	1	3	4	3
SEL-01	Alternatives evaluation and select decision	Process	.	4	4	4	4*
SEL-02	Technology and licensor selection	Process	.	4	4	4	4*
SEL-03	Utility and offsite basis	Process	.	2	3	4	2
SAF-02	Global flare load study	Safety	.	1	3	4	0
SAF-01	Preliminary HAZOP	Safety	.	.	4	4	4
SAF-04	Siting study, QRA, F&G mapping	Safety	.	.	4	4	0
SAF-03	Formal HAZOP and LOPA workshop	Safety	.	.	.	4	0

ID	Activity	Discipline	FEL 1	FEL 2	FEL 3	DD	Now
MEC-01	Mechanical design calculations	Mechanical	.	1	3	4	3
MEC-02	Equipment datasheets	Mechanical	.	1	3	4	3
MEC-03	Equipment GA drawings	Mechanical	.	.	2	4	2
VEN-01	Vendor enquiries and vendor data	Mechanical	.	.	2	4	0
PRO-01	Requisitions and technical bid evaluations	Procurement	.	.	3	4	0
FEL-03	Site screening and selection	Layout	2	4	4	4	4*
LAY-01	Plot plan and equipment layout	Layout	1	2	3	4	3
LAY-02	Sections and 3D layout model	Layout	.	1	3	4	3
LAY-03	3D model reviews and constructability	Layout	.	.	2	4	0
PIP-01	Piping material classes	Piping	.	1	3	4	3
PIP-02	Piping routing and 3D model	Piping	.	.	2	4	2
PIP-03	Pipe stress and supports (screening)	Piping	.	.	2	4	1
PIP-04	Isometrics and piping MTO	Piping	.	.	2	4	2
PIP-05	Detailed pipe stress (CAESAR II)	Piping	.	.	1	4	0
CIV-01	Geotechnical, civil and structural design	Civil	.	1	2	4	0
CIV-02	Foundation and structural steel detailing	Civil	.	.	.	4	0
ICS-01	Control philosophy and schemes	I&C	.	1	3	4	3
ICS-07	ICS architecture	I&C	.	1	3	4	3
ICS-02	SIF list and SIL determination (LOPA)	I&C	.	.	2	4	1
ICS-03	Cause and effect matrix	I&C	.	.	2	4	2
ICS-04	Control valve sizing	I&C	.	.	2	4	2
ICS-05	I/O list	I&C	.	.	2	4	2
ICS-06	Loop diagrams	I&C	.	.	1	4	1
ICS-08	SIL verification and SRS	I&C	.	.	.	4	0

ID	Activity	Discipline	FEL 1	FEL 2	FEL 3	DD	Now
ICS-09	Instrument installation design	I&C	.	.	.	4	0
ELE-01	Electrical load list	Electrical	.	1	3	4	3
ELE-03	Electrical sizing, single-line diagrams	Electrical	.	1	3	4	3
ELE-02	Hazardous area classification	Electrical	.	.	3	4	3
ELE-04	Cable schedule	Electrical	.	.	2	4	2
ELE-05	Power system studies and protection	Electrical	.	.	1	4	0
ELE-06	Electrical layouts	Electrical	.	.	.	4	0
CON-01	Constructability, modularisation and path of construction	Construction	.	1	3	4	0
CON-02	Work packaging and system boundaries	Construction	.	.	1	4	0
FEL-04	Class 5 estimate and economics	Cost	4	4	4	4	4*
CST-01	Class 4 cost estimate	Cost	.	4	4	4	4
CST-02	Class 3 cost estimate	Cost	.	.	4	4	0
CST-03	Control estimate	Cost	.	.	.	4	0

* assumed: decided in the design basis rather than developed in this package.

5 Link maturity rules

- By default, taking an activity to level L needs each of its inputs at level L.
- Studies, estimates and HAZOPs need early inputs only. Their cap: FEL-02 2, FEL-03 2, FEL-04 1, SEL-01 2, SEL-02 2, PRC-05 2, PRC-07 3, SAF-01 2, SAF-03 3, SAF-04 2, VEN-01 3, CST-01 2, CST-02 2, CST-03 3.
- Single-link exceptions:
 - PRC-00 → PRC-01: at level 1 needs nothing, at level 2 needs 1 (calibrated cut points and efficiencies)
 - SEL-01 → PRC-01: at level 1 needs nothing, at level 2 needs nothing (selected configuration)
 - SEL-01 → CST-01: at level 4 needs 4 (selected scope)
 - SEL-02 → SEL-01: at level 4 needs 4 (selected technology)
 - SEL-01 → PRJ-05: at level 1 needs nothing (selected scope)
 - PRC-04 → PRC-06: at level 1 needs nothing, at level 2 needs 1 (principal control loops)
 - PIP-03 → CIV-01: at level 1 needs nothing (support loads)
 - ICS-05 → ICS-07: at level 1 needs nothing, at level 3 needs 2 (I/O counts)
 - MEC-01 → CST-01: at level 4 needs 1 (shell weights)

- Iteration loops become forward links between levels, so the plan has no cycles:
- PRC-02 reaches level 3 after MEC-01 reaches level 2 (mechanical holds re-rate sizing (C-201 bed-1 diameter, heater box length))
- LAY-01 reaches level 3 after MEC-01 reaches level 2 (calculated dimensions and weights update footprints and spacing)
- PRC-02 reaches level 4 after PIP-02 reaches level 3 (routed lengths update pump and line hydraulics)
- LAY-01 reaches level 4 after PIP-02 reaches level 3 (rack width and clashes move equipment)
- PRC-08 reaches level 4 after ICS-04 reaches level 3 (control valve sizes and reducers shown on the P&IDs)
- PRC-08 reaches level 3 after SAF-01 reaches level 4 (HAZOP recommendations revise the P&IDs)
- ELE-03 reaches level 4 after ELE-04 reaches level 3 (actual cable impedances re-run the motor-starting voltage-dip check)
- ELE-01 reaches level 4 after ICS-07 reaches level 3 (ICS cabinet and UPS loads added to the load list)
- PRC-03 reaches level 4 after SAF-02 reaches level 3 (flare back-pressure changes PSV type and size)
- PRC-08 reaches level 4 after SAF-03 reaches level 4 (formal HAZOP actions revise the P&IDs)
- LAY-01 reaches level 3 after SAF-04 reaches level 4 (siting study moves occupied buildings and equipment)
- MEC-01 reaches level 4 after VEN-01 reaches level 4 (certified vendor dimensions, weights and nozzles)
- LAY-01 reaches level 4 after VEN-01 reaches level 4 (vendor package footprints)
- ELE-01 reaches level 4 after VEN-01 reaches level 4 (certified motor ratings)
- ICS-05 reaches level 4 after VEN-01 reaches level 4 (vendor package I/O)

6 Gate readiness of this package

Gate	Needed	Met	Assumed	Short
Gate 1 · opportunity confirmed	13	8	4	1
Gate 2 · concept selected	34	22	6	6
Gate 3 · final investment decision	54	29	6	19
Gate 4 · issued for construction	61	5	6	50

Gate 1 · opportunity confirmed: short

- PRJ-05 Execution plan and contracting strategy: at 0 not started, needs 1 concept

Gate 2 · concept selected: short

- PRJ-05 Execution plan and contracting strategy: at 0 not started, needs 2 preliminary
- SEL-04 Permitting and environmental basis: at 0 not started, needs 2 preliminary
- PRC-00 Rigorous simulation and model calibration: at 0 not started, needs 1 concept
- SAF-02 Global flare load study: at 0 not started, needs 1 concept
- CIV-01 Geotechnical, civil and structural design: at 0 not started, needs 1 concept
- CON-01 Constructability, modularisation and path of construction: at 0 not started, needs 1 concept

Gate 3 · final investment decision: short

- PRJ-05 Execution plan and contracting strategy: at 0 not started, needs 4 final
- SEL-04 Permitting and environmental basis: at 0 not started, needs 3 defined
- PRC-01 Heat and material balance: at 2 preliminary, needs 3 defined

- PRJ-02 Crude assay characterisation: at 2 preliminary, needs 4 final
- PRC-00 Rigorous simulation and model calibration: at 0 not started, needs 4 final
- SEL-03 Utility and offsite basis: at 2 preliminary, needs 3 defined
- SAF-02 Global flare load study: at 0 not started, needs 3 defined
- SAF-04 Siting study, QRA, F&G mapping: at 0 not started, needs 4 final
- VEN-01 Vendor enquiries and vendor data: at 0 not started, needs 2 preliminary
- PRO-01 Requisitions and technical bid evaluations: at 0 not started, needs 3 defined
- LAY-03 3D model reviews and constructability: at 0 not started, needs 2 preliminary
- PIP-03 Pipe stress and supports (screening): at 1 concept, needs 2 preliminary
- PIP-05 Detailed pipe stress (CAESAR II): at 0 not started, needs 1 concept
- CIV-01 Geotechnical, civil and structural design: at 0 not started, needs 2 preliminary
- ICS-02 SIF list and SIL determination (LOPA): at 1 concept, needs 2 preliminary
- ELE-05 Power system studies and protection: at 0 not started, needs 1 concept
- CON-01 Constructability, modularisation and path of construction: at 0 not started, needs 3 defined
- CON-02 Work packaging and system boundaries: at 0 not started, needs 1 concept
- CST-02 Class 3 cost estimate: at 0 not started, needs 4 final

7 Steps by phase

Each step takes an activity to a level. Steps are in network order; "ready" means every input is already at the level the step needs.

FEL 1 · Appraise

23 steps: 22 done, 1 ready, 0 blocked.

Order	ID	Activity	To level	Today	Waiting on
1	FEL-01	Business case, capacity and product slate	1 Concept	done	-
1	PRJ-03	Document register and portal	1 Concept	done	-
2	FEL-01	Business case, capacity and product slate	2 Preliminary	done	-
2	FEL-03	Site screening and selection	1 Concept	done	-
2	PRJ-01	Design basis (BOD)	1 Concept	done	-
2	PRJ-05	Execution plan and contracting strategy	1 Concept	ready	-
3	FEL-01	Business case, capacity and product slate	3 Defined	done	-
3	FEL-02	Configuration options and screening	1 Concept	done	-
3	FEL-03	Site screening and selection	2 Preliminary	done	-
3	PRJ-01	Design basis (BOD)	2 Preliminary	done	-
3	PRJ-02	Crude assay characterisation	1 Concept	done	-
4	FEL-02	Configuration options and screening	2 Preliminary	done	-

Order	ID	Activity	To level	Today	Waiting on
4	PRC-01	Heat and material balance	1 Concept	done	-
5	FEL-02	Configuration options and screening	3 Defined	done	-
5	PRC-02	Process equipment sizing	1 Concept	done	-
5	PRC-05	Block flow diagram	1 Concept	done	-
6	FEL-02	Configuration options and screening	4 Final	done	-
6	LAY-01	Plot plan and equipment layout	1 Concept	done	-
6	PRC-06	Process flow diagrams	1 Concept	done	-
7	FEL-04	Class 5 estimate and economics	1 Concept	done	-
8	FEL-04	Class 5 estimate and economics	2 Preliminary	done	-
9	FEL-04	Class 5 estimate and economics	3 Defined	done	-
10	FEL-04	Class 5 estimate and economics	4 Final	done	-

FEL 2 - Select

45 steps: 38 done, 4 ready, 3 blocked.

Order	ID	Activity	To level	Today	Waiting on
2	PRJ-03	Document register and portal	2 Preliminary	done	-
4	FEL-01	Business case, capacity and product slate	4 Final	done	-
4	FEL-03	Site screening and selection	3 Defined	done	-
4	PRC-00	Rigorous simulation and model calibration	1 Concept	ready	-
4	PRJ-01	Design basis (BOD)	3 Defined	done	-
4	PRJ-02	Crude assay characterisation	2 Preliminary	done	-
4	SEL-02	Technology and licensor selection	1 Concept	done	-
5	FEL-03	Site screening and selection	4 Final	done	-
5	PIP-01	Piping material classes	1 Concept	done	-
5	PRC-01	Heat and material balance	2 Preliminary	done	PRC-00 at 1
5	SEL-01	Alternatives evaluation and select decision	1 Concept	done	-
5	SEL-02	Technology and licensor selection	2 Preliminary	done	-
5	SEL-03	Utility and offsite basis	1 Concept	done	-
6	ELE-01	Electrical load list	1 Concept	done	-
6	PRC-02	Process equipment sizing	2 Preliminary	done	-

Order	ID	Activity	To level	Today	Waiting on
6	PRC-03	Relief load and PSV sizing (unit)	1 Concept	done	-
6	PRC-04	Control loops and SIF definition	1 Concept	done	-
6	PRC-05	Block flow diagram	2 Preliminary	done	-
6	SEL-01	Alternatives evaluation and select decision	2 Preliminary	done	-
6	SEL-02	Technology and licensor selection	3 Defined	done	-
6	SEL-03	Utility and offsite basis	2 Preliminary	done	-
7	ELE-03	Electrical sizing, single-line diagrams	1 Concept	done	-
7	ICS-01	Control philosophy and schemes	1 Concept	done	-
7	LAY-01	Plot plan and equipment layout	2 Preliminary	done	-
7	LAY-02	Sections and 3D layout model	1 Concept	done	-
7	MEC-01	Mechanical design calculations	1 Concept	done	-
7	PRC-05	Block flow diagram	3 Defined	done	-
7	PRC-06	Process flow diagrams	2 Preliminary	done	-
7	PRC-07	Process design report	1 Concept	done	-
7	PRC-08	P&IDs, line list, instrument index	1 Concept	done	-
7	PRJ-05	Execution plan and contracting strategy	2 Preliminary	blocked	PRJ-05 at 1
7	SAF-02	Global flare load study	1 Concept	ready	-
7	SEL-01	Alternatives evaluation and select decision	3 Defined	done	-
7	SEL-02	Technology and licensor selection	4 Final	done	-
7	SEL-04	Permitting and environmental basis	1 Concept	ready	-
8	CIV-01	Geotechnical, civil and structural design	1 Concept	ready	-
8	CON-01	Constructability, modularisation and path of construction	1 Concept	blocked	PRJ-05 at 1
8	CST-01	Class 4 cost estimate	1 Concept	done	-
8	ICS-07	ICS architecture	1 Concept	done	-
8	MEC-02	Equipment datasheets	1 Concept	done	-
8	SEL-01	Alternatives evaluation and select decision	4 Final	done	-
8	SEL-04	Permitting and environmental basis	2 Preliminary	blocked	SEL-04 at 1
9	CST-01	Class 4 cost estimate	2 Preliminary	done	-
10	CST-01	Class 4 cost estimate	3 Defined	done	-
11	CST-01	Class 4 cost estimate	4 Final	done	-

FEL 3 · Define (FEED)

90 steps: 55 done, 4 ready, 31 blocked.

Order	ID	Activity	To level	Today	Waiting on
3	PRJ-03	Document register and portal	3 Defined	done	-
5	PRC-00	Rigorous simulation and model calibration	2 Preliminary	blocked	PRC-00 at 1
5	PRJ-01	Design basis (BOD)	4 Final	done	-
5	PRJ-02	Crude assay characterisation	3 Defined	ready	-
6	PIP-01	Piping material classes	2 Preliminary	done	-
6	PRC-00	Rigorous simulation and model calibration	3 Defined	blocked	PRJ-02 at 3, PRC-00 at 2
6	PRJ-02	Crude assay characterisation	4 Final	blocked	PRJ-02 at 3
7	ELE-01	Electrical load list	2 Preliminary	done	-
7	ELE-02	Hazardous area classification	1 Concept	done	-
7	PRC-00	Rigorous simulation and model calibration	4 Final	blocked	PRJ-02 at 4, PRC-00 at 3
7	PRC-03	Relief load and PSV sizing (unit)	2 Preliminary	done	-
7	PRC-04	Control loops and SIF definition	2 Preliminary	done	-
8	ELE-02	Hazardous area classification	2 Preliminary	done	-
8	ELE-03	Electrical sizing, single-line diagrams	2 Preliminary	done	-
8	ELE-04	Cable schedule	1 Concept	done	-
8	ICS-01	Control philosophy and schemes	2 Preliminary	done	-
8	ICS-04	Control valve sizing	1 Concept	done	-
8	ICS-05	I/O list	1 Concept	done	-
8	LAY-02	Sections and 3D layout model	2 Preliminary	done	-
8	MEC-01	Mechanical design calculations	2 Preliminary	done	-
8	MEC-03	Equipment GA drawings	1 Concept	done	-
8	PIP-02	Piping routing and 3D model	1 Concept	done	-
8	PRC-01	Heat and material balance	3 Defined	blocked	PRJ-02 at 3, PRC-00 at 3
8	PRC-05	Block flow diagram	4 Final	done	-
8	PRC-07	Process design report	2 Preliminary	done	-
8	PRC-08	P&IDs, line list, instrument index	2 Preliminary	done	-
8	PRJ-05	Execution plan and contracting strategy	3 Defined	blocked	PRJ-05 at 2
8	SAF-01	Preliminary HAZOP	1 Concept	done	-
8	SAF-02	Global flare load study	2 Preliminary	blocked	SAF-02 at 1

Order	ID	Activity	To level	Today	Waiting on
9	CON-01	Constructability, modularisation and path of construction	2 Preliminary	blocked	PRJ-05 at 2, CON-01 at 1
9	ELE-04	Cable schedule	2 Preliminary	done	-
9	ICS-02	SIF list and SIL determination (LOPA)	1 Concept	done	-
9	ICS-04	Control valve sizing	2 Preliminary	done	-
9	ICS-05	I/O list	2 Preliminary	done	-
9	ICS-06	Loop diagrams	1 Concept	done	-
9	LAY-03	3D model reviews and constructability	1 Concept	blocked	CIV-01 at 1
9	MEC-02	Equipment datasheets	2 Preliminary	done	-
9	MEC-03	Equipment GA drawings	2 Preliminary	done	-
9	PIP-01	Piping material classes	3 Defined	done	PRC-01 at 3
9	PIP-02	Piping routing and 3D model	2 Preliminary	done	-
9	PIP-03	Pipe stress and supports (screening)	1 Concept	done	-
9	PIP-04	Isometrics and piping MTO	1 Concept	done	-
9	PRC-02	Process equipment sizing	3 Defined	done	PRC-01 at 3
9	PRJ-05	Execution plan and contracting strategy	4 Final	blocked	PRJ-05 at 3
9	PRO-01	Requisitions and technical bid evaluations	1 Concept	blocked	PRJ-05 at 1
9	SAF-01	Preliminary HAZOP	2 Preliminary	done	-
9	SAF-04	Siting study, QRA, F&G mapping	1 Concept	ready	-
9	SEL-03	Utility and offsite basis	3 Defined	blocked	PRC-01 at 3
10	CON-02	Work packaging and system boundaries	1 Concept	blocked	CON-01 at 1
10	ELE-01	Electrical load list	3 Defined	done	-
10	ICS-02	SIF list and SIL determination (LOPA)	2 Preliminary	ready	-
10	ICS-03	Cause and effect matrix	1 Concept	done	-
10	ICS-07	ICS architecture	2 Preliminary	done	-
10	PIP-03	Pipe stress and supports (screening)	2 Preliminary	ready	-
10	PIP-04	Isometrics and piping MTO	2 Preliminary	done	-
10	PRC-03	Relief load and PSV sizing (unit)	3 Defined	done	PRC-01 at 3
10	PRC-04	Control loops and SIF definition	3 Defined	done	PRC-01 at 3
10	PRO-01	Requisitions and technical bid evaluations	2 Preliminary	blocked	PRJ-05 at 2, PRO-01 at 1
10	SAF-01	Preliminary HAZOP	3 Defined	done	-

Order	ID	Activity	To level	Today	Waiting on
10	SAF-04	Siting study, QRA, F&G mapping	2 Preliminary	blocked	SAF-04 at 1
10	VEN-01	Vendor enquiries and vendor data	1 Concept	blocked	PRO-01 at 1
11	CIV-01	Geotechnical, civil and structural design	2 Preliminary	blocked	PIP-03 at 2, CIV-01 at 1
11	CST-02	Class 3 cost estimate	1 Concept	blocked	VEN-01 at 1, CIV-01 at 1
11	ELE-03	Electrical sizing, single-line diagrams	3 Defined	done	-
11	ELE-05	Power system studies and protection	1 Concept	blocked	VEN-01 at 1
11	ICS-01	Control philosophy and schemes	3 Defined	done	PRC-01 at 3
11	ICS-03	Cause and effect matrix	2 Preliminary	done	ICS-02 at 2
11	MEC-01	Mechanical design calculations	3 Defined	done	PRC-01 at 3
11	PIP-05	Detailed pipe stress (CAESAR II)	1 Concept	blocked	VEN-01 at 1
11	PRC-06	Process flow diagrams	3 Defined	done	PRC-01 at 3
11	PRC-07	Process design report	3 Defined	done	PRC-01 at 3
11	SAF-01	Preliminary HAZOP	4 Final	done	-
11	SAF-02	Global flare load study	3 Defined	blocked	SAF-02 at 2
11	SAF-04	Siting study, QRA, F&G mapping	3 Defined	blocked	SAF-04 at 2
11	VEN-01	Vendor enquiries and vendor data	2 Preliminary	blocked	PRO-01 at 2, VEN-01 at 1
12	CST-02	Class 3 cost estimate	2 Preliminary	blocked	VEN-01 at 2, CIV-01 at 2, CST-02 at 1
12	ICS-07	ICS architecture	3 Defined	done	-
12	LAY-03	3D model reviews and constructability	2 Preliminary	blocked	CIV-01 at 2, LAY-03 at 1
12	MEC-02	Equipment datasheets	3 Defined	done	-
12	PRC-07	Process design report	4 Final	done	PRC-01 at 3
12	SAF-04	Siting study, QRA, F&G mapping	4 Final	blocked	SAF-04 at 3
13	CST-02	Class 3 cost estimate	3 Defined	blocked	VEN-01 at 2, CIV-01 at 2, CST-02 at 2
13	LAY-01	Plot plan and equipment layout	3 Defined	done	PRC-01 at 3, SAF-04 at 4
13	PRO-01	Requisitions and technical bid evaluations	3 Defined	blocked	PRJ-05 at 3, PRO-01 at 2
14	CST-02	Class 3 cost estimate	4 Final	blocked	VEN-01 at 2, CIV-01 at 2, CST-02 at 3
14	ELE-02	Hazardous area classification	3 Defined	done	-
14	LAY-02	Sections and 3D layout model	3 Defined	done	-

Order	ID	Activity	To level	Today	Waiting on
14	PRC-08	P&IDs, line list, instrument index	3 Defined	done	-
14	SEL-04	Permitting and environmental basis	3 Defined	blocked	PRC-01 at 3, SEL-04 at 2
15	CON-01	Constructability, modularisation and path of construction	3 Defined	blocked	PRJ-05 at 3, CON-01 at 2

Detailed design

86 steps: 0 done, 7 ready, 79 blocked.

Order	ID	Activity	To level	Today	Waiting on
9	ELE-06	Electrical layouts	1 Concept	ready	-
9	PRC-01	Heat and material balance	4 Final	blocked	PRJ-02 at 4, PRC-00 at 4, PRC-01 at 3
10	ELE-06	Electrical layouts	2 Preliminary	blocked	ELE-06 at 1
10	ICS-06	Loop diagrams	2 Preliminary	ready	-
10	PIP-01	Piping material classes	4 Final	blocked	PRC-01 at 4
10	SAF-03	Formal HAZOP and LOPA workshop	1 Concept	ready	-
10	SEL-03	Utility and offsite basis	4 Final	blocked	PRC-01 at 4, SEL-03 at 3
11	CON-02	Work packaging and system boundaries	2 Preliminary	blocked	CON-01 at 2, CON-02 at 1
11	ICS-02	SIF list and SIL determination (LOPA)	3 Defined	blocked	ICS-02 at 2
11	ICS-08	SIL verification and SRS	1 Concept	blocked	SAF-03 at 1, VEN-01 at 1
11	ICS-09	Instrument installation design	1 Concept	blocked	VEN-01 at 1
11	SAF-03	Formal HAZOP and LOPA workshop	2 Preliminary	blocked	ICS-02 at 2, SAF-03 at 1
12	CIV-02	Foundation and structural steel detailing	1 Concept	blocked	CIV-01 at 1, VEN-01 at 1, PIP-05 at 1
12	CST-03	Control estimate	1 Concept	blocked	CST-02 at 1, PRO-01 at 1, CIV-01 at 1
12	ELE-05	Power system studies and protection	2 Preliminary	blocked	VEN-01 at 2, ELE-05 at 1
12	ICS-03	Cause and effect matrix	3 Defined	blocked	ICS-02 at 3
12	ICS-08	SIL verification and SRS	2 Preliminary	blocked	ICS-02 at 2, SAF-03 at 2, VEN-01 at 2, ICS-08 at 1
12	ICS-09	Instrument installation design	2 Preliminary	blocked	VEN-01 at 2, ICS-09 at 1
12	MEC-03	Equipment GA drawings	3 Defined	ready	-
12	PIP-05	Detailed pipe stress (CAESAR II)	2 Preliminary	blocked	PIP-03 at 2, VEN-01 at 2, PIP-05 at 1
13	CIV-02	Foundation and structural steel detailing	2 Preliminary	blocked	CIV-01 at 2, VEN-01 at 2, PIP-05 at 2, CIV-02 at 1

Order	ID	Activity	To level	Today	Waiting on
13	CST-03	Control estimate	2 Preliminary	blocked	CST-02 at 2, PRO-01 at 2, CIV-01 at 2, CST-03 at 1
13	PRJ-04	PE review, seal and IFC issue	1 Concept	blocked	SAF-03 at 1, ICS-08 at 1, PIP-05 at 1, ELE-05 at 1, CST-02 at 1, LAY-03 at 1, SAF-04 at 1, SAF-02 at 1, CIV-02 at 1, ICS-09 at 1, ELE-06 at 1, CON-02 at 1, PRO-01 at 1
14	PRJ-04	PE review, seal and IFC issue	2 Preliminary	blocked	SAF-03 at 2, ICS-08 at 2, PIP-05 at 2, ELE-05 at 2, CST-02 at 2, LAY-03 at 2, SAF-04 at 2, SAF-02 at 2, CIV-02 at 2, ICS-09 at 2, ELE-06 at 2, CON-02 at 2, PRO-01 at 2, PRJ-04 at 1
14	VEN-01	Vendor enquiries and vendor data	3 Defined	blocked	PRO-01 at 3, VEN-01 at 2
15	ELE-04	Cable schedule	3 Defined	ready	-
15	ELE-05	Power system studies and protection	3 Defined	blocked	VEN-01 at 3, ELE-05 at 2
15	ICS-04	Control valve sizing	3 Defined	blocked	PRC-01 at 3
15	ICS-05	I/O list	3 Defined	ready	-
15	PIP-02	Piping routing and 3D model	3 Defined	ready	-
15	SAF-03	Formal HAZOP and LOPA workshop	3 Defined	blocked	ICS-02 at 3, SAF-03 at 2
15	VEN-01	Vendor enquiries and vendor data	4 Final	blocked	PRO-01 at 3, VEN-01 at 3
16	ELE-06	Electrical layouts	3 Defined	blocked	ELE-04 at 3, PIP-02 at 3, ELE-06 at 2
16	ICS-06	Loop diagrams	3 Defined	blocked	ICS-05 at 3, ICS-06 at 2
16	ICS-08	SIL verification and SRS	3 Defined	blocked	ICS-02 at 3, SAF-03 at 3, VEN-01 at 3, ICS-08 at 2
16	ICS-09	Instrument installation design	3 Defined	blocked	ICS-05 at 3, PIP-02 at 3, VEN-01 at 3, ICS-09 at 2
16	PIP-03	Pipe stress and supports (screening)	3 Defined	blocked	PIP-02 at 3, PIP-03 at 2
16	PIP-04	Isometrics and piping MTO	3 Defined	blocked	PIP-02 at 3
16	PRC-02	Process equipment sizing	4 Final	blocked	PRC-01 at 4, PIP-02 at 3
16	SAF-03	Formal HAZOP and LOPA workshop	4 Final	blocked	ICS-02 at 3, SAF-03 at 3
17	CIV-01	Geotechnical, civil and structural design	3 Defined	blocked	PIP-03 at 3, CIV-01 at 2
17	CON-02	Work packaging and system boundaries	3 Defined	blocked	CON-01 at 3, PIP-04 at 3, CON-02 at 2
17	ELE-01	Electrical load list	4 Final	blocked	PRC-02 at 4, VEN-01 at 4

Order	ID	Activity	To level	Today	Waiting on
17	LAY-01	Plot plan and equipment layout	4 Final	blocked	PRC-02 at 4, PRC-01 at 4, PIP-02 at 3, SAF-04 at 4, VEN-01 at 4
17	PIP-05	Detailed pipe stress (CAESAR II)	3 Defined	blocked	PIP-03 at 3, VEN-01 at 3, PIP-05 at 2
17	PRC-03	Relief load and PSV sizing (unit)	4 Final	blocked	PRC-01 at 4, PRC-02 at 4, SAF-02 at 3
17	PRC-04	Control loops and SIF definition	4 Final	blocked	PRC-01 at 4, PRC-02 at 4
18	CIV-02	Foundation and structural steel detailing	3 Defined	blocked	CIV-01 at 3, VEN-01 at 3, PIP-05 at 3, CIV-02 at 2
18	CST-03	Control estimate	3 Defined	blocked	CST-02 at 3, PRO-01 at 3, PIP-04 at 3, CIV-01 at 3, CST-03 at 2
18	ELE-02	Hazardous area classification	4 Final	blocked	LAY-01 at 4, PRC-02 at 4, PRC-04 at 4
18	ELE-03	Electrical sizing, single-line diagrams	4 Final	blocked	ELE-01 at 4, ELE-04 at 3
18	ICS-01	Control philosophy and schemes	4 Final	blocked	PRC-04 at 4, PRC-01 at 4, PRC-02 at 4
18	ICS-02	SIF list and SIL determination (LOPA)	4 Final	blocked	PRC-04 at 4, ICS-02 at 3
18	LAY-02	Sections and 3D layout model	4 Final	blocked	LAY-01 at 4, PRC-02 at 4
18	LAY-03	3D model reviews and constructability	3 Defined	blocked	PIP-02 at 3, CIV-01 at 3, LAY-03 at 2
18	MEC-01	Mechanical design calculations	4 Final	blocked	PRC-02 at 4, PRC-01 at 4, PRC-03 at 4, VEN-01 at 4
18	PRC-06	Process flow diagrams	4 Final	blocked	PRC-01 at 4, PRC-02 at 4, PRC-04 at 4
18	SAF-02	Global flare load study	4 Final	blocked	PRC-03 at 4, SAF-02 at 3
18	SEL-04	Permitting and environmental basis	4 Final	blocked	PRC-01 at 4, LAY-01 at 4, SEL-04 at 3
19	CON-01	Constructability, modularisation and path of construction	4 Final	blocked	LAY-01 at 4, LAY-02 at 4, PRJ-05 at 4, CON-01 at 3
19	CST-03	Control estimate	4 Final	blocked	CST-02 at 3, PRO-01 at 3, PIP-04 at 3, CIV-01 at 3, CST-03 at 3
19	ELE-04	Cable schedule	4 Final	blocked	ELE-01 at 4, ELE-03 at 4, LAY-01 at 4, ELE-02 at 4, ELE-04 at 3
19	ELE-05	Power system studies and protection	4 Final	blocked	ELE-03 at 4, VEN-01 at 4, ELE-05 at 3
19	ICS-03	Cause and effect matrix	4 Final	blocked	PRC-04 at 4, PRC-03 at 4, PRC-02 at 4, ICS-02 at 4, ICS-03 at 3

Order	ID	Activity	To level	Today	Waiting on
19	ICS-08	SIL verification and SRS	4 Final	blocked	ICS-02 at 4, SAF-03 at 4, VEN-01 at 4, ICS-08 at 3
19	MEC-02	Equipment datasheets	4 Final	blocked	MEC-01 at 4, PRC-02 at 4, PRC-03 at 4
19	MEC-03	Equipment GA drawings	4 Final	blocked	MEC-01 at 4, MEC-03 at 3
19	PRC-08	P&IDs, line list, instrument index	4 Final	blocked	PRC-02 at 4, PRC-03 at 4, PRC-04 at 4, PRC-06 at 4, PIP-01 at 4, LAY-01 at 4, ICS-04 at 3, SAF-03 at 4
19	PRJ-04	PE review, seal and IFC issue	3 Defined	blocked	SAF-03 at 3, ICS-08 at 3, PIP-05 at 3, ELE-05 at 3, CST-02 at 3, LAY-03 at 3, SAF-04 at 3, SAF-02 at 3, CIV-02 at 3, ICS-09 at 3, ELE-06 at 3, CON-02 at 3, PRO-01 at 3, PRJ-04 at 2
20	ICS-04	Control valve sizing	4 Final	blocked	PRC-01 at 4, PRC-02 at 4, PRC-04 at 4, PRC-08 at 4, ICS-04 at 3
20	ICS-05	I/O list	4 Final	blocked	PRC-08 at 4, PRC-04 at 4, PRC-02 at 4, LAY-01 at 4, VEN-01 at 4, ICS-05 at 3
20	PIP-02	Piping routing and 3D model	4 Final	blocked	PRC-08 at 4, LAY-01 at 4, PRC-03 at 4, MEC-01 at 4, PIP-02 at 3
20	PRO-01	Requisitions and technical bid evaluations	4 Final	blocked	MEC-02 at 4, PRJ-05 at 4, PRO-01 at 3
21	ELE-06	Electrical layouts	4 Final	blocked	ELE-04 at 4, LAY-02 at 4, PIP-02 at 4, ELE-06 at 3
21	ICS-06	Loop diagrams	4 Final	blocked	ICS-05 at 4, ICS-06 at 3
21	ICS-07	ICS architecture	4 Final	blocked	ICS-05 at 4, ELE-03 at 4
21	ICS-09	Instrument installation design	4 Final	blocked	ICS-05 at 4, PIP-02 at 4, VEN-01 at 4, ICS-09 at 3
21	PIP-03	Pipe stress and supports (screening)	4 Final	blocked	PIP-02 at 4, MEC-01 at 4, PIP-03 at 3
21	PIP-04	Isometrics and piping MTO	4 Final	blocked	PIP-02 at 4, PIP-01 at 4, PIP-04 at 3
22	CIV-01	Geotechnical, civil and structural design	4 Final	blocked	LAY-01 at 4, MEC-01 at 4, PIP-03 at 4, CIV-01 at 3
22	CON-02	Work packaging and system boundaries	4 Final	blocked	CON-01 at 4, PRC-08 at 4, PIP-04 at 4, CON-02 at 3
22	PIP-05	Detailed pipe stress (CAESAR II)	4 Final	blocked	PIP-03 at 4, VEN-01 at 4, PIP-05 at 3

Order	ID	Activity	To level	Today	Waiting on
22	PRJ-03	Document register and portal	4 Final	blocked	PIP-04 at 4, ICS-06 at 4, ICS-07 at 4, ELE-04 at 4, MEC-03 at 4, LAY-02 at 4, ICS-01 at 4, ICS-03 at 4, ICS-04 at 4, MEC-02 at 4, PIP-03 at 4
23	CIV-02	Foundation and structural steel detailing	4 Final	blocked	CIV-01 at 4, VEN-01 at 4, PIP-05 at 4, CIV-02 at 3
23	LAY-03	3D model reviews and constructability	4 Final	blocked	PIP-02 at 4, LAY-02 at 4, CIV-01 at 4, LAY-03 at 3
24	PRJ-04	PE review, seal and IFC issue	4 Final	blocked	PRJ-03 at 4, SAF-03 at 4, ICS-08 at 4, PIP-05 at 4, ELE-05 at 4, CST-02 at 4, LAY-03 at 4, SAF-04 at 4, SAF-02 at 4, CIV-02 at 4, ICS-09 at 4, ELE-06 at 4, CON-02 at 4, PRO-01 at 4, PRJ-04 at 3

8 Consistency check

- none: every gate target is supported by its inputs at the same gate, and the step network has no cycles

9 Limits

- Targets are typical values for a refinery process unit and should be agreed with the owner's gate criteria (for example CII PDRI or IPA FEL index) before use.
- One level per activity: an activity whose deliverables mature at different rates (for example equipment datasheets for long-lead and bulk items) is shown at the level of most of its deliverables.
- "Achieved" is taken from the package status: issued = the FEL 3 target, provisional = the FEL 2 target.
- Construction, commissioning and handover phases are not modelled.